PROJECT PROGRESS LOG

THERMOMETER

Team Members

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Roll Numbers

EE26BTECH11211
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1 Building circuit on Breadboard

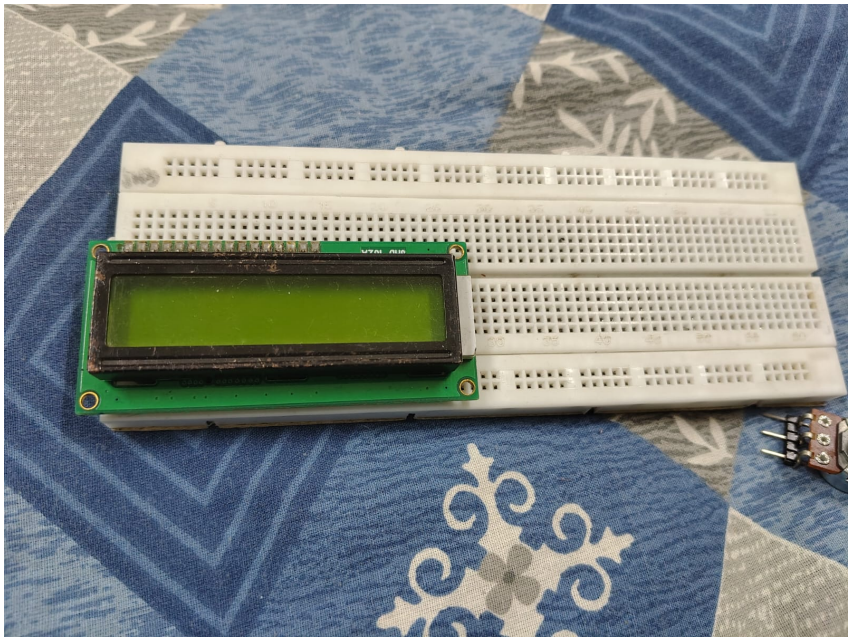


Figure 1: LCD on breadboard

1.1 Workdone (Figure: 1)

We first tested the apparatus we received from the lab. We wanted to test the whole setup on a breadboard first. Then we placed the LCD on the breadboard as shown in Figure 1

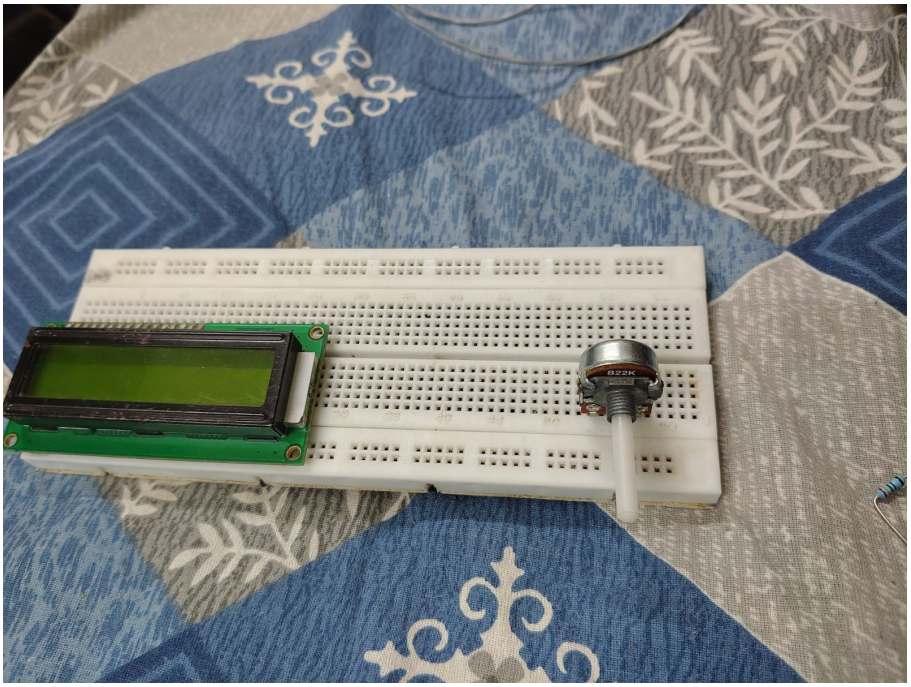


Figure 2: Potentiometer on breadboard

1.2 Workdone (Figure: 2)

Then we attached the potentiometer on the breadboard, besides the LCD display, as shown in Figure 2

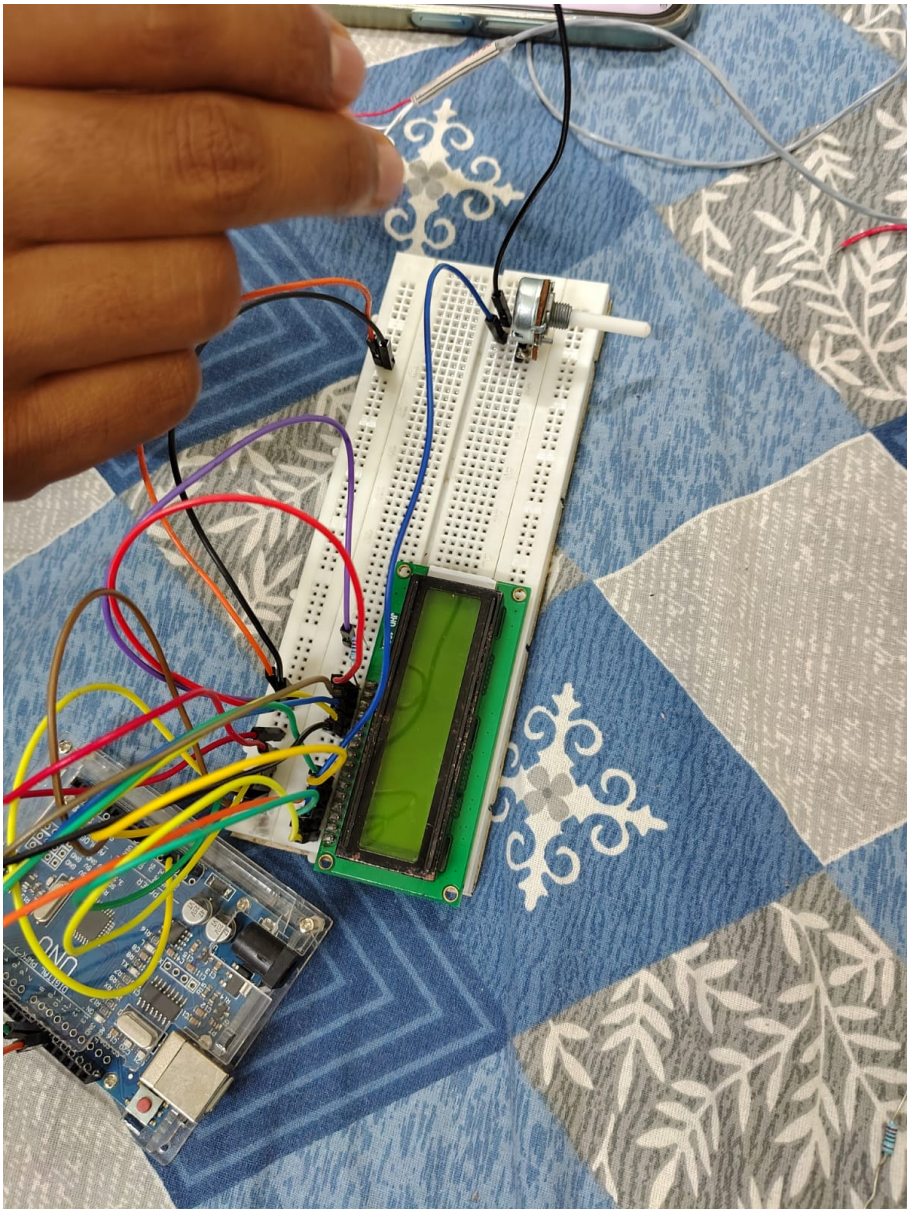


Figure 3: Wiring the breadboard and the ardiuno with jumper wires

1.3 Workdone (Figure: 3)

After placing the LCD and the potentiometer on the breadboard, we made connections using jumper wires. We refer to pt100 repository for the connections to be made between the Ardiuno, LCD and Potentiometer. As shown in Figure 3 we made the connections between the apparatus and connected Pt100 to the circuit.

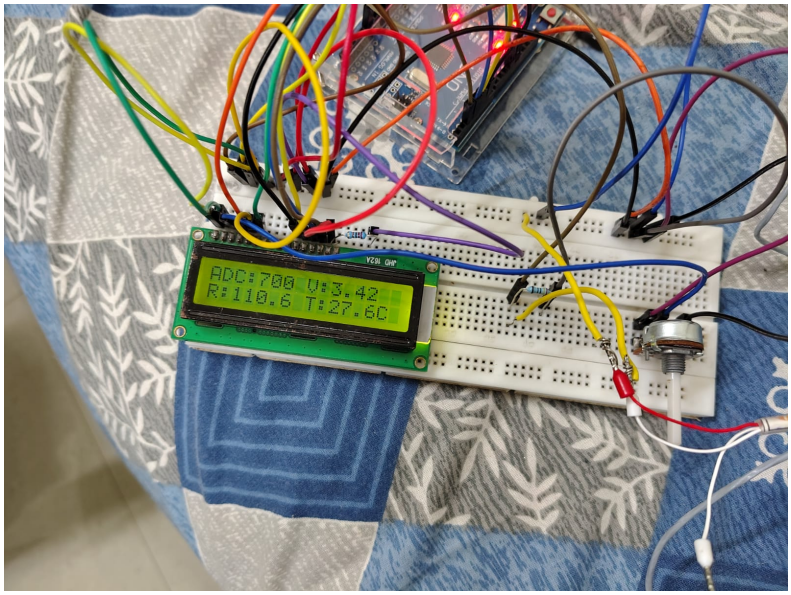


Figure 4: Readings of the thermometer at room temperature

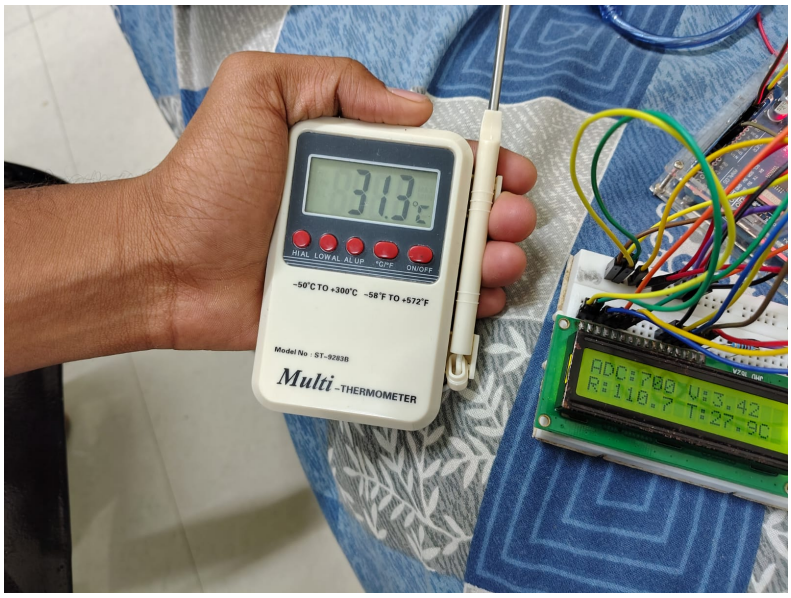


Figure 5: Readings of the lab thermometer at room temperature

1.4 Observation (Figure: 4,5)

After the whole connection was completed on the breadboard, we gave power to the Ardiuno. The LCD started to glow. We then enter the code into the Arduino. Then we observed the values as seen in Figuure 4. It indicated room temperature. We measured room temperature with an industrial thermometer and the thermometer we made .We observed about 27.6°**C. The actual room temperature was** 31°**C.**

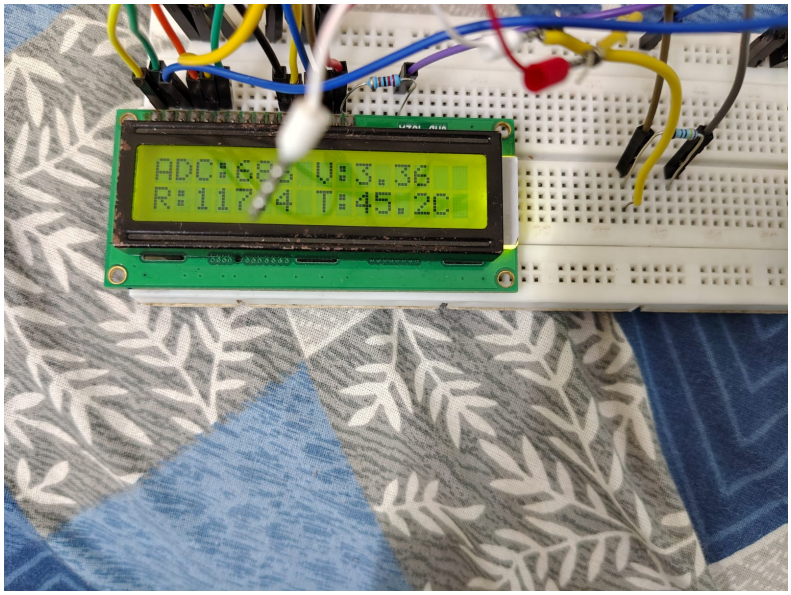


Figure 6: Reading of our thermometer in hot water



Figure 7: Reading of lab thermometer in hot water

1.5 Observation (Figure: 6,7)

We observe the readings of both thermometers in hot water. We got 45.2°C as our temperature reading and 52.8°C as the lab thermometer reading.Then we realized that there is a slight error in obtaining the temperature reading we obtained from our temperature. We observed this in both the room temperature reading and the hot water temperature reading.

2 Building the circuit on the Perfboard

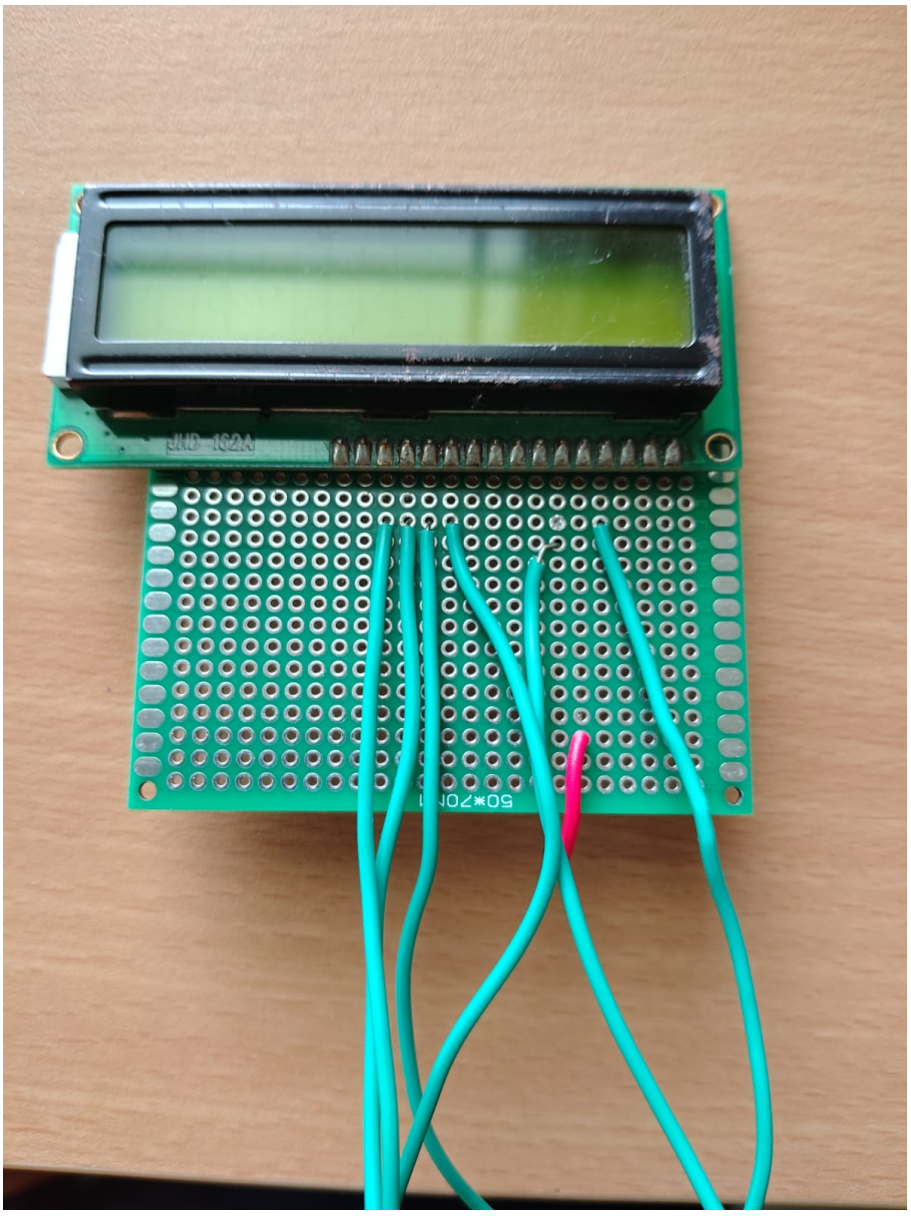


Figure 8: Arrangemet of LCD and wires on the Perfboard

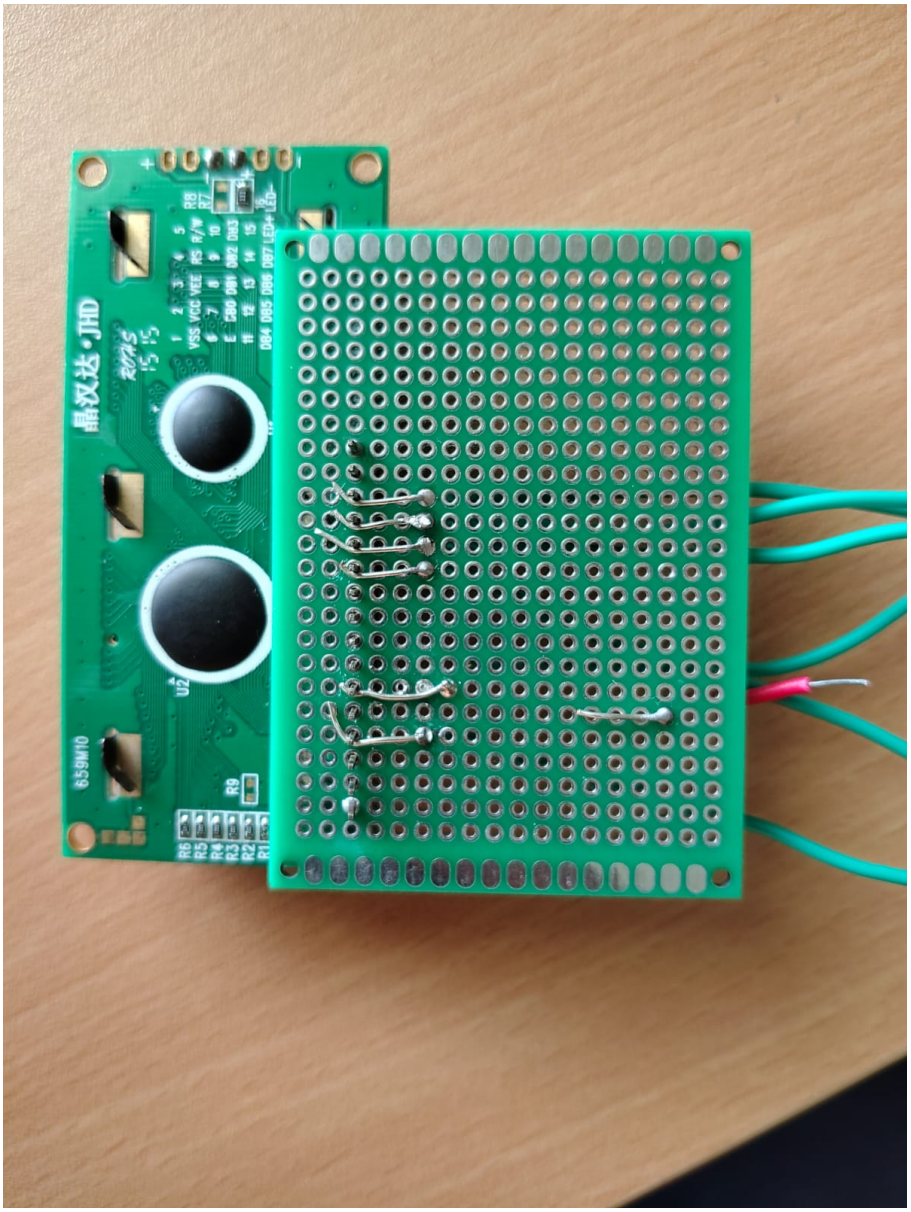


Figure 9: Other side of the Perfboard

2.1 Workdone(Figure: 8,9)

We connected the LCD to the perfboard and soldered it firmly to the perfboard. Then we connected the wires and soldered them firmly as well, As shown in figure 8 and Figure 9

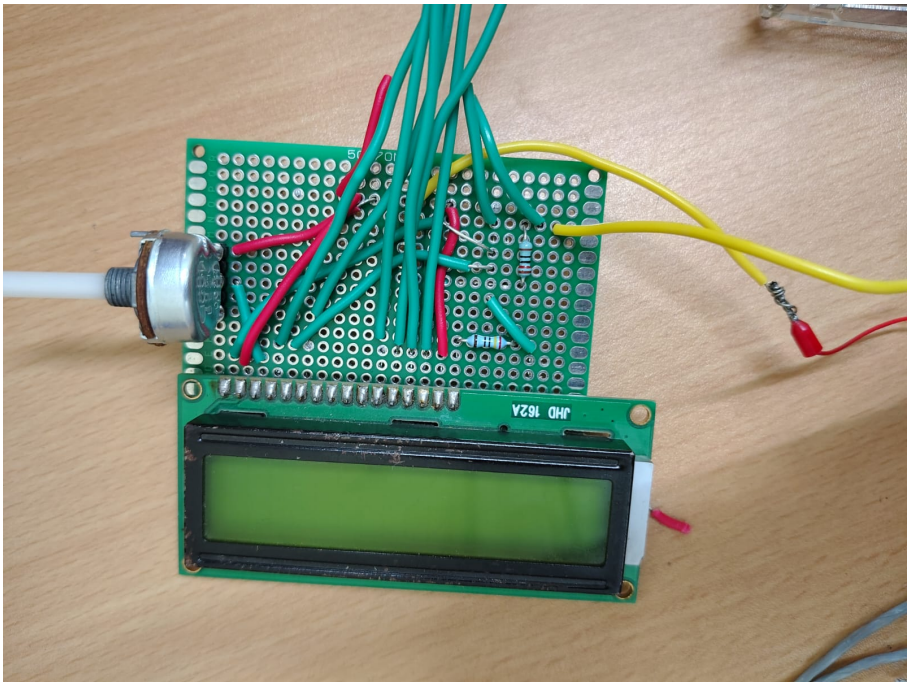


Figure 10: The perfboard after all the connections are made as we done on the breadboard

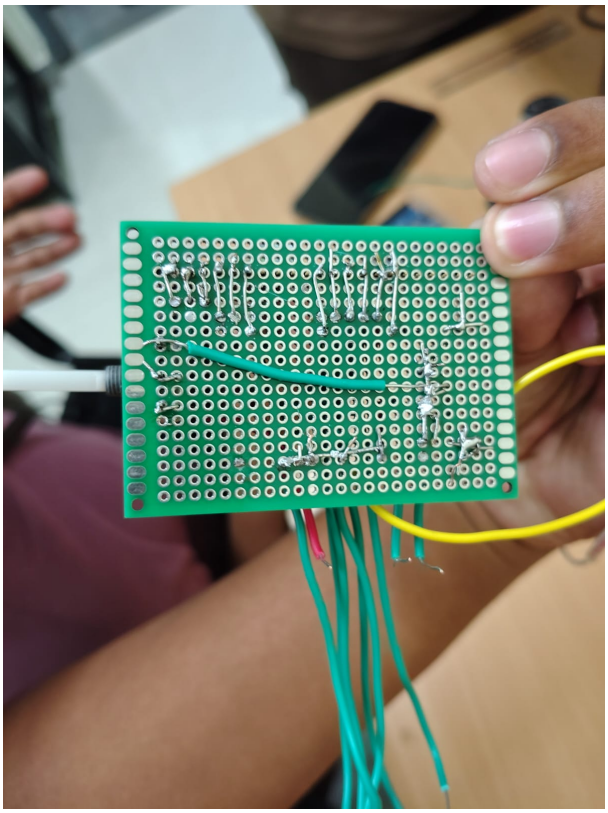


Figure 11: The other side of the perfboard after all the connections are made and soldered firmly.

2.2 Workdone(Figure: 10,11)

We made connections on the perfboard as we did on the breadboard. Then we soldered the wires and the resistors on the perfboard.. We did not solder the wires directly to the LCD. we just left them touching to the LCD, As we can see in Figure 11.

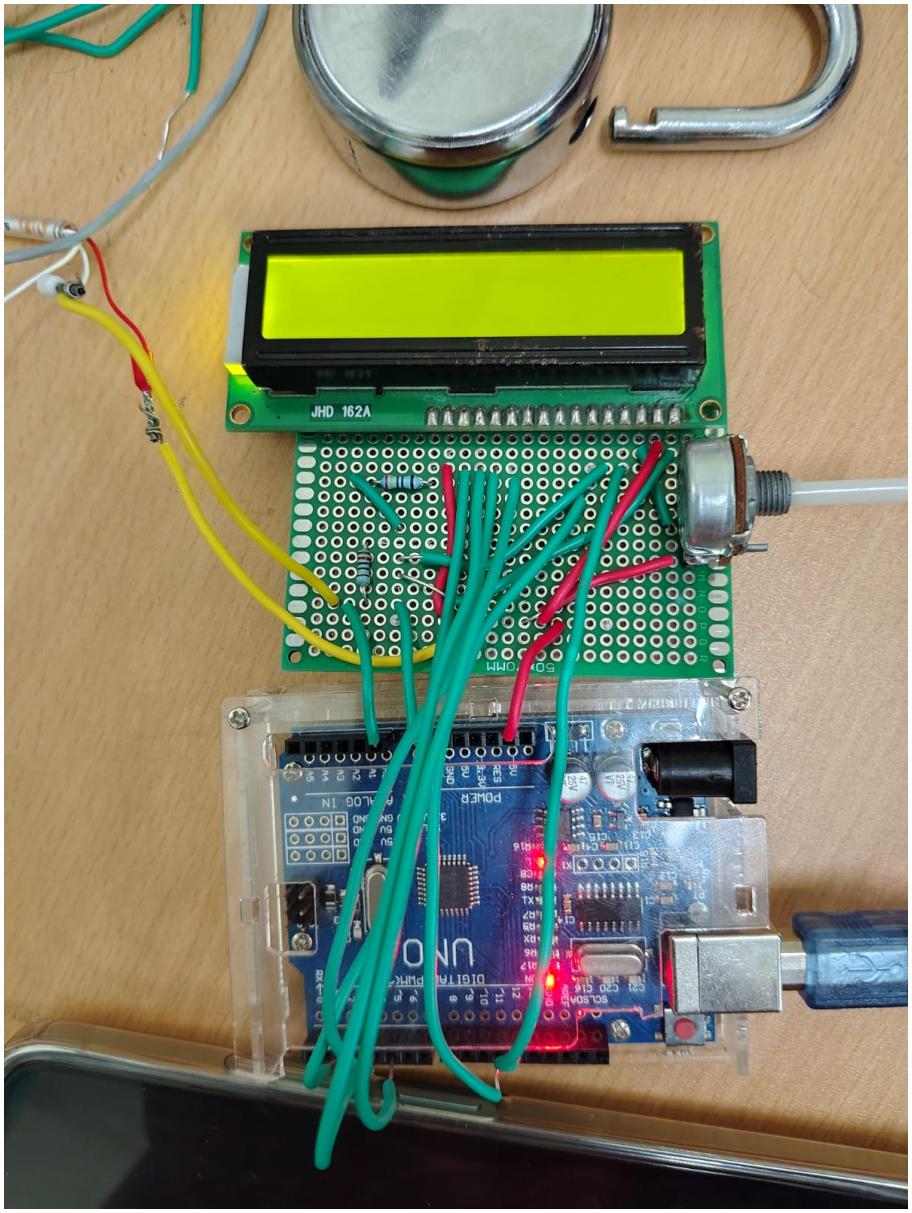


Figure 12: Connecting to the Ardiuno and giving power

2.3 Observation (Figure: 12)

After connecting the whole circuit to the Ardiuno, we gave power to the whole thing. We observed that after giving power the LCD started to glow, but did not print or show anything.

2.4 Problem (Figure: 12)

We faced a problem with the LCD. It did not display anything. It just glows.

2.5 What we did

We thought that there were some loose connections, as we did not solder the wires to the LCD. We then gently re-soldered the LCD on the perfboard and removed it from the perfboard.

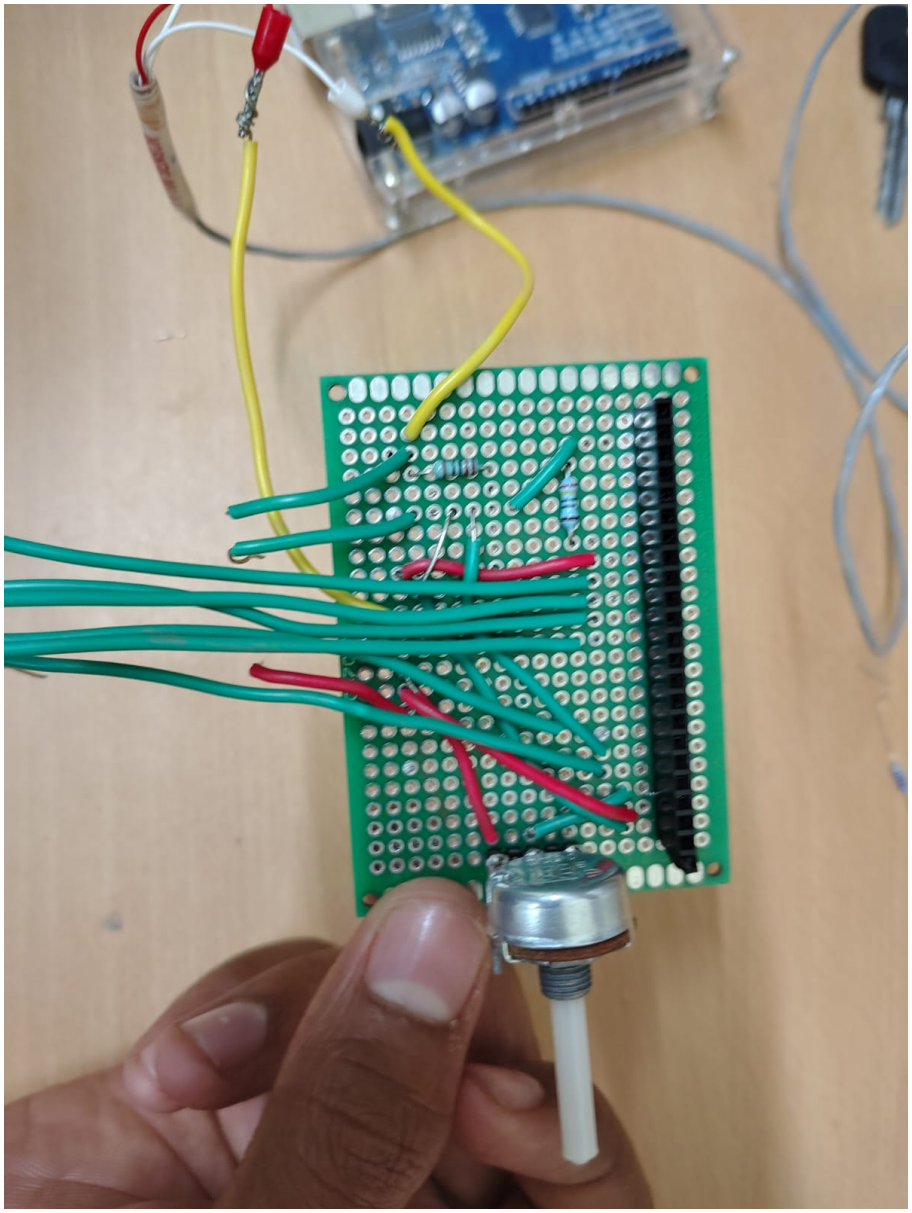


Figure 13: Adding header pins to the perfboard

2.6 Workdone(Figure: 13)

After gently re-soldering the LCD, we added the header ins to the perfboard. Then we solder it to the perfboard. Then we soldered the header pins to the wires that were left touching the LCD.

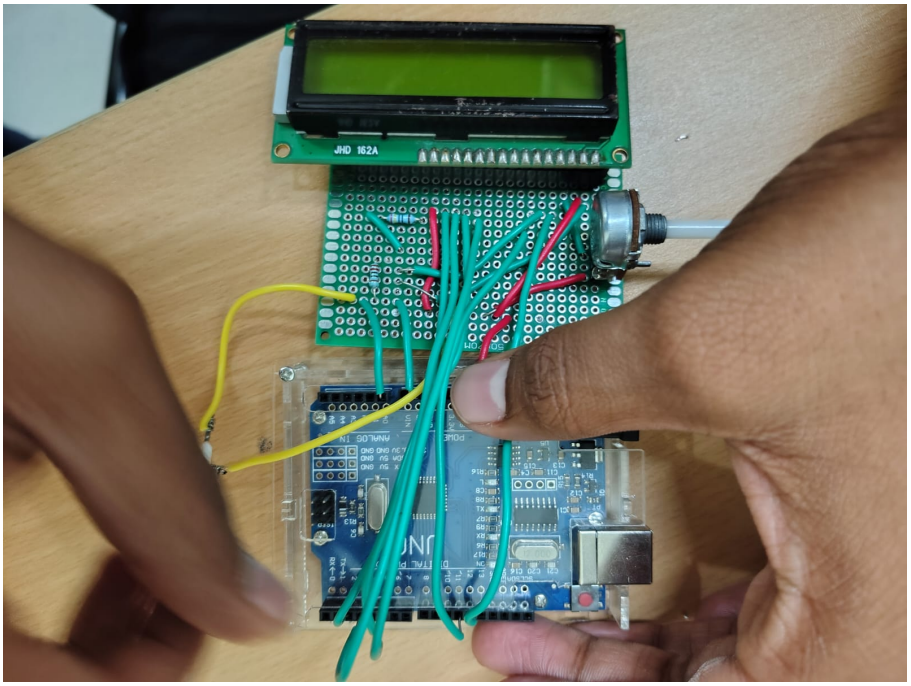


Figure 14: After adding header pins, reconnecting the circuit

2.7 Workdone (Figure: 14)

We then attached the LCD to the perfboard with the help of soldered header pins on the perfboard.After attaching the LCD to the perfboard we made connections to the ardiuno and suppliying it power.

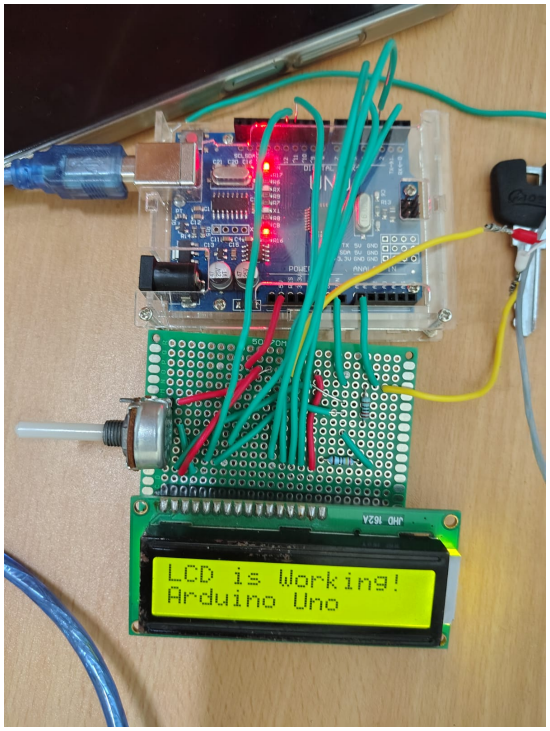


Figure 15: Testing of LCD

2.8 Observation (Figure: 15)

After supplying power to the ardiuno, we checked if the LCD display was working properly or not. It was confirmed that it is working as it displayed what we wanted to print. Hence LCD was working.after verifying the working of LCD, we changed the code to measure the temperature.

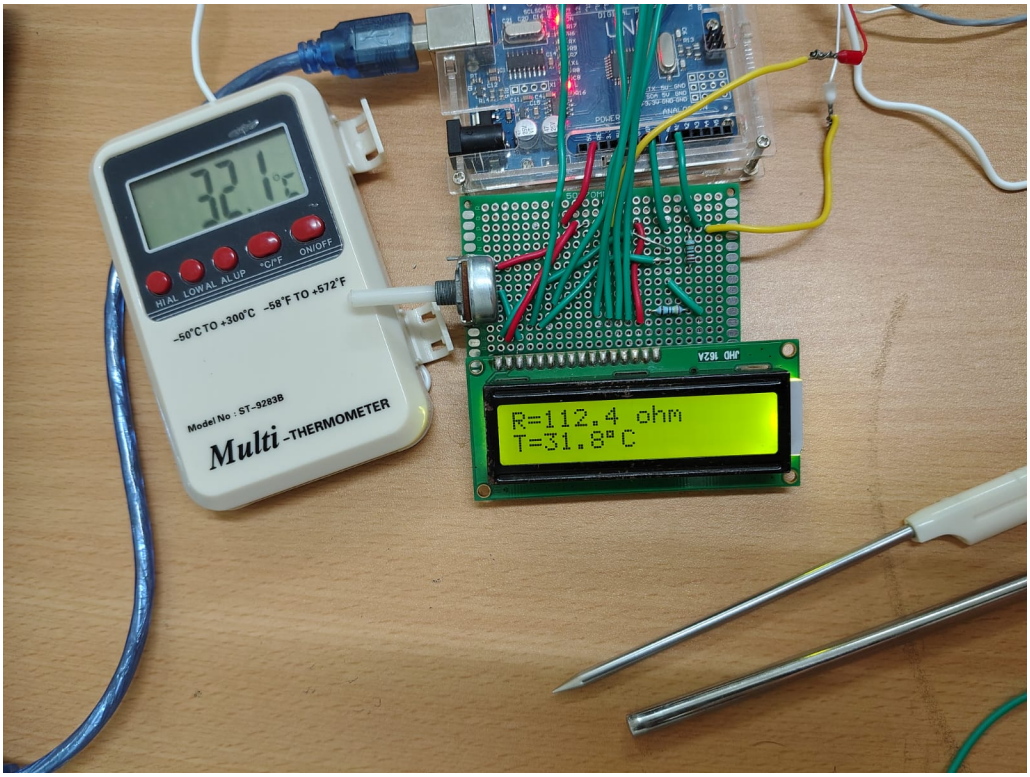


Figure 16: Readings of both thermometers at room temperature

2.9 Observation (Figure: 16)

At room temperature we measured the temperature with both the temperatures. The lab thermometer shows about 32.1°C and our temperature was showing about 31.8°C. As there was error in our room temperature measurement, we measured the temperature of hot water in a kettle.

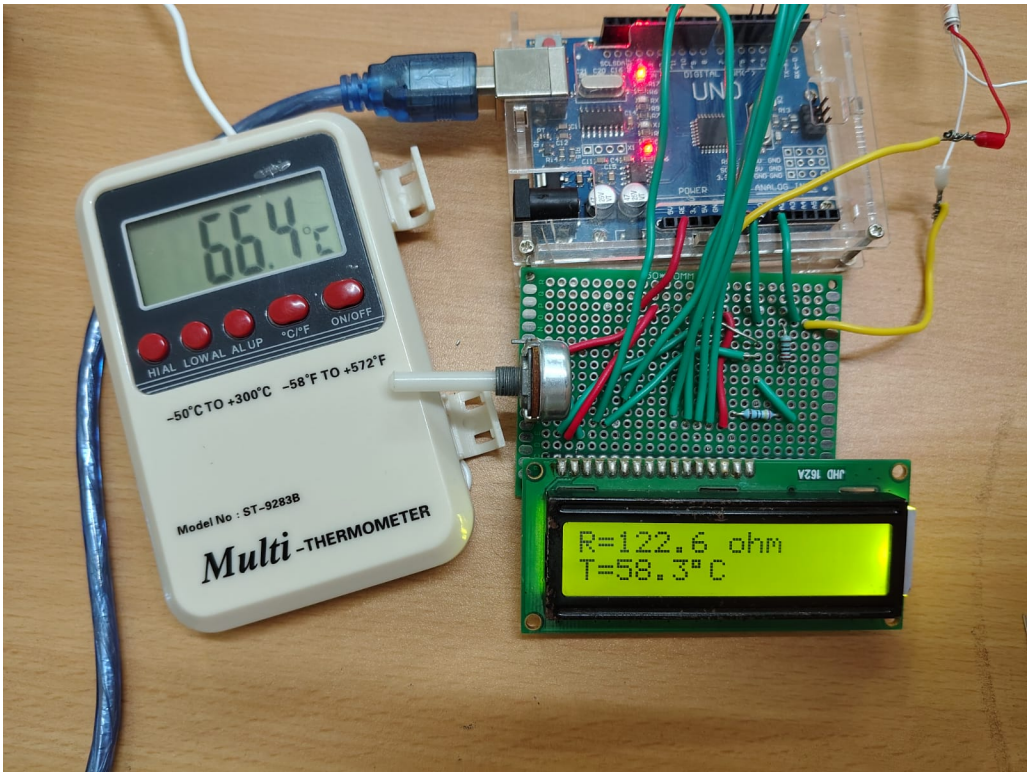


Figure 17: Readings of both thermometers of hot water

2.10 Observation (Figure: 17)

After taking the readings of room temperature with both thermometers. We took readings oh hot water in a kettle of both thermometers. The lab thermometer shows 66.4°C where as our thermometer shows 58.3°C. We observed

2.11 Conclusion

We concluded that our thermometer has an error in measuring the temperature. It is not as acurate as the lab thermometer.We modified our code.

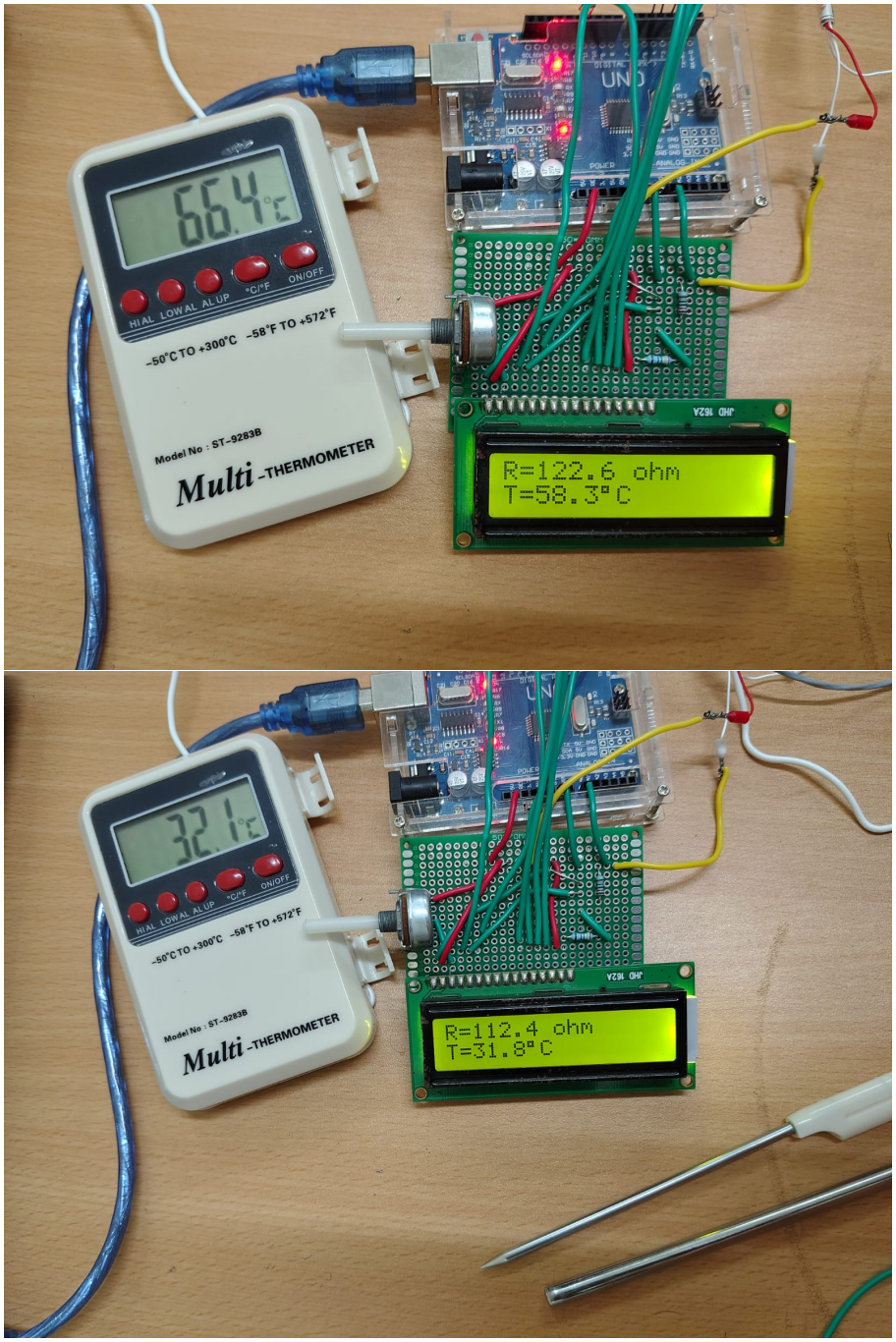


Figure 18: Verified values required for calibration

2.12 Workdone (Figure: 18)

We compared the values of voltage and temperature needed for calibration and modified the code of the ardiuno.

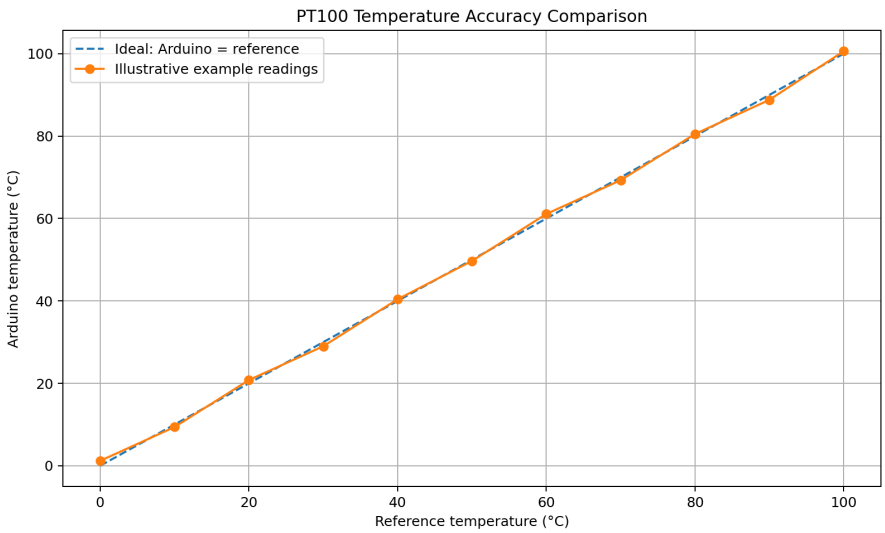


Figure 19: Accuracy graph

$$T = aX^2 + b^2 + c$$

We used this equation for calibrating. We used the values of V from the Figure 18.